Cadmiel Avendaño HW 11

1) A first order.

$A \rightarrow B$ $\quad k = 2\ min^{-1}$ $\quad 0 < t < 5\,min,\ 1\sin(Bt) = E(t)$

$t > 5\,min \quad 0 = E(t)$

RTD function = $A\sin(Bt)$.

$$\int_0^5 E(t) = 1$$

a) $\tau = \int_0^\infty t\,E(t)\,dt$

$$\tau = \int_0^5 t\,E(t)\,dt + \cancel{\int_5^\infty t(E(t))\,dt} = \tau = \int_0^5 t\,A\sin(Bt)\,dt.$$

$$t\left(-\frac{A}{B}\cos Bt\right) + \frac{A}{B}\int \cos(Bt)\,dt = +\frac{A}{B^2}\sin(Bt)$$

$$A\left(\frac{1}{B^2}\sin(Bt) - \frac{t}{B}\cos(Bt)\right)\Bigg|_0^5 = \left(\frac{1}{B^2}\sin(5B) - \frac{5}{B}\cos(5B)\right) - (\cancel{\sin 0} - 0)$$

$$\boxed{\tau = A\left[\frac{1}{B^2}\sin(B5) - \frac{5}{B}\cos(B5)\right]}$$

$$A\int_0^5 \sin Bt = 1 \Rightarrow \frac{A}{B}\left[\cos(5B) + \cos(0)\right] = 1 \Rightarrow 1 - \cos(5B) = \frac{B}{A}$$

$\cos 5B = 1 - \frac{B}{A}$ $\qquad B = \frac{\pi}{5} \therefore$ $\qquad \cos\pi = \left(1 - \frac{B}{A}\right)$ $\quad \frac{B}{A} = 2,\ A = \frac{B}{2}$

$A = \frac{\pi}{10}$

$$\frac{\pi}{10}\left[\frac{5^2}{\pi^2}\cancel{\sin(\pi)}^{\,0} - \frac{25}{\pi}\overset{(-1)}{\cos(\pi)}\right] = \cancel{0} + \frac{25}{\cancel{\pi}}\frac{\cancel{\pi}}{10} = \frac{25}{10} = \boxed{2.5 = \tau \text{ min}}$$

b). $\int_0^5 E(t)\cdot X(t)\,dt$, $X(t) = 1-e^{-kt}$

$$\int_0^5 A\sin(Bt)\left(1-e^{-kt}\right)dt = \frac{\pi}{10}\int_0^5 \sin\left(\frac{\pi}{5}t\right)\left(1-e^{-2t}\right)dt$$

$$\frac{\pi}{10}\left[\left(1-e^{-2t}\right)\left(-\frac{5}{\pi}\cos\left(\frac{\pi}{5}t\right)\right) - \int 2e^{-2t}\left(-\frac{5}{\pi}\cos\left(\frac{\pi}{5}t\right)\right)dt\right]$$

$$= \frac{\pi}{10}\left[\left[\frac{10}{\pi}\frac{e^{-t2}}{\left(\frac{\pi}{5}\right)^2+4}\left(2\cos\left(\frac{\pi}{5}t\right) - \frac{\pi}{5}\sin\left(\frac{\pi}{5}t\right)\right)\right] - \left(\frac{5}{\pi}\left(1-e^{-2t}\right)\cos\left(\frac{\pi}{5}t\right)\right)\right]$$

$$= \left(\frac{\cancel{e^{-10}}}{\left(\frac{\pi}{5}\right)^2+4}\left(2\overset{(-1)}{\cos\pi} - \frac{\pi}{5}\overset{0}{\sin\pi}\right)\right) + \left[\frac{\overset{0}{\cancel{e^{-10}}}-1}{2}\overset{-1}{\cancel{\cos\pi}}\right]$$

$$-\frac{1}{\left(\frac{\pi}{5}\right)^2+4}(2) + 0 \Rightarrow \frac{1-e^{-10}}{2} + \frac{e^{-10}}{\left(\frac{\pi}{5}\right)^2+4}(-2) - \frac{2}{\frac{\pi^2}{5}+4}$$

$$.49 - 2.066\times10^{-5} - 4.6 = \boxed{X = .0455}$$

d) P at $\frac{5}{3}$ & $\frac{5}{2}$ min

$$= \int_{5/3}^{5/2} E(t)\,dt = \frac{\pi}{10}\int \sin\left(\frac{\pi}{5}t\right)dt = -\frac{\pi}{10}\frac{5}{\pi}\frac{1}{2}\cos\frac{\pi}{5}t\Big|_{5/3}^{5/2} =$$

$$-\frac{1}{2}\cos\frac{\pi}{2} - \cos\frac{\pi}{3} = \frac{1}{2}\cos\frac{\pi}{3} = \boxed{.25}$$

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4. CSTR, 2nd order

$C_{A0} = .5 \frac{mol}{L}$

$X = .80$ in 400 seconds.

$$\text{final } C_A = .5 \cdot (1-.80) = .1 \frac{mol}{L} = C_A \text{ final}$$

$$\frac{C_A}{C_{A0}} = \frac{1}{1 C_{A0}kt} = \frac{1}{1+(.5)k} = \frac{.1}{.05} \Rightarrow \underline{t = 8 \text{ min}}$$

$$\frac{C_A}{C_{A0}} = \frac{-1+\sqrt{4C_{A0}kt+1}}{2C_{A0}kt} = \frac{-1+\sqrt{1+4(\;)(400)}}{2(.5)\,400}$$

$$C_A = .068 \cdot .5 = C_A = .034 \frac{mols}{L}$$

$$X_A = \frac{.5 - .034}{.5} = \boxed{X_A = 93\%}$$